NAGAVALLI MAREEDU

Senior AI/ML Engineer | Gen AI Full Stack Developer

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PROFESSIONAL SUMMARY

Senior AI/ML Engineer with 5+ years of experience architecting and shipping production-grade Generative AI systems, LLM pipelines, and full-stack ML applications at scale. Deep expertise across the end-to-end model lifecycle—from fine-tuning open-source LLMs (LLaMA 3, GPT-4) using LoRA/QLoRA and designing RAG architectures with LangChain and LlamaIndex, to containerized microservice deployment on AWS, GCP, and Azure via Docker, Kubernetes, and CI/CD automation. Proven ability to bridge research and engineering: translating business requirements into measurable AI outcomes, embedding MLOps best practices (MLflow, DVC, Prometheus), and delivering human-aligned GenAI experiences across cross-functional teams. Adept at driving latency reduction, cost optimization, and accuracy improvements in high-throughput inference environments.

TECHNICAL SKILLS

Languages & Frameworks: Python, JavaScript/TypeScript, SQL, FastAPI, Flask, Node.js, React, Next.js, Tailwind CSS, REST APIs, GraphQL

AI/ML & GenAI: LLMs (GPT-4, Claude, Gemini, LLaMA 3), RAG Pipelines, GraphRAG, Agentic AI, Multi-Agent Systems, AI Agents, Prompt Engineering, Fine-Tuning (LoRA/QLoRA), Tool Calling, Function Calling, LLM Evaluation, AI Guardrails, Semantic Search, Hybrid Search, Re-ranking, Embeddings, Hugging Face Transformers, LangChain, LlamaIndex, LangGraph, Sentence Transformers, Scikit-learn, XGBoost, PyTorch, TensorFlow

MLOps & Model Deployment: MLflow, DVC, Triton Inference Server, Kubernetes, Docker, GitHub Actions CI/CD, OpenTelemetry, Prometheus, Grafana, LangSmith, Model Monitoring, Drift Detection, A/B Testing

Data Engineering: ETL/ELT Pipelines, Apache Airflow, PostgreSQL, MongoDB, BigQuery, Redis, Celery, FAISS, Pinecone, ChromaDB, Weaviate, Vector Databases, Chunking & Indexing Strategies

Cloud Platforms: AWS (S3, ECS, Lambda, CloudWatch, SageMaker, Bedrock), GCP (Vertex AI, BigQuery, Cloud Vision, Gemini), Azure ML, Azure OpenAI, Terraform

Visualization & Reporting: Streamlit, Gradio, Plotly, Dash, Tableau, Power BI

Responsible AI & Governance: Responsible AI, Explainable AI (XAI), Human-in-the-Loop AI, AI Governance, Model Risk Management, HIPAA Compliance, PII/PHI Handling

PROFESSIONAL EXPERIENCE

Sr. AI Engineer | **FedEx**, United States *July 2025 – Present*

- **Designed an enterprise LLM platform** supporting chat assistants, summarization, and document Q&A—reducing average analyst research time by ~45% by automating knowledge-base retrieval across internal document stores using LangChain, FAISS, and GPT-4.
- **Engineered end-to-end RAG pipelines** with LangChain and LlamaIndex, integrating FAISS vector stores and recursive-splitter chunking strategies, improving contextual retrieval precision by 30% and materially cutting hallucination rates on domain Q&A benchmarks.
- **Fine-tuned LLaMA 3 and GPT variants** using LoRA/QLoRA on domain-specific logistics datasets, achieving a 12% improvement in F1-score on intent classification while reducing inference cost by 20% versus full-model API calls.
- **Built and deployed 8+ GenAI microservices** via REST and GraphQL APIs, containerized with Docker and orchestrated on Kubernetes; applied token streaming and Redis caching to sustain <200 ms P95 latency under 500+ concurrent users.
- **Shipped React/Next.js frontends** with WebSocket streaming for real-time GenAI interaction, reducing perceived response latency by 60% and lifting user engagement scores by 35% in A/B testing.
- **Established MLOps CI/CD pipelines** using GitHub Actions for automated model training, evaluation gating, and blue-green deployment to Kubernetes—cutting release cycle time from 2 weeks to 3 days.

- **Instrumented model observability** with OpenTelemetry, Prometheus, and custom logging, enabling real-time drift detection and reducing mean time to incident resolution (MTTR) by 50%.
- **Applied advanced prompt engineering** (chain-of-thought, few-shot, system prompt scaffolding) to reduce model output drift by 40% and align responses with enterprise compliance and safety requirements.

AI Engineer | **State Farm**, United States *January 2025 – June 2025*

- **Designed and deployed a production RAG-based document Q&A system** enabling natural-language querying across 50,000+ unstructured insurance documents, reducing manual document review time by 60% for internal underwriting teams using LangChain, LlamaIndex, and OpenAI GPT-4.
- **Built a FastAPI backend** containerized with Docker and deployed on AWS ECS with auto-scaling; leveraged S3 for document storage and CloudWatch for observability, achieving 99.9% uptime over a 6-month production window.
- **Fine-tuned sentence-transformer embeddings** on domain-specific insurance text with recursive chunking and HNSW indexing in FAISS, improving semantic retrieval MRR@10 by 22% versus baseline OpenAI Ada embeddings.
- **Automated vector index maintenance** via PostgreSQL triggers for embedding refresh on document updates, eliminating manual re-indexing and ensuring index freshness with zero additional engineering overhead.
- **Implemented structured output generation** using OpenAI function calling and JSON schema validation, enabling downstream workflow integrations and reducing post-processing engineering effort by 35%.
- **Conducted multi-provider LLM benchmarking** (OpenAI, Anthropic, Cohere) across latency, accuracy, and cost dimensions, informing a provider selection that reduced monthly inference spend by $8K while maintaining quality parity.
- **Deployed LangSmith tracing and Prometheus metrics** for full auditability of LLM interactions, enabling compliance reporting and reducing debugging time for edge-case query failures by 70%.

ML Engineer | **Infosys**, India *August 2021 – December 2023*

- **Delivered end-to-end ML pipelines** for real-time classification, recommendation, and NLP tasks using PyTorch, XGBoost, and scikit-learn on Azure Kubernetes Service (AKS)—powering models that processed 1M+ daily inference requests at <50 ms average latency.
- **Designed and optimized data ingestion pipelines** using Apache Airflow and BigQuery, reducing feature engineering runtime by 55% through parallelized ETL orchestration and query optimization across multi-terabyte datasets.
- **Implemented MLflow and DVC for full model lifecycle management** across 15+ experiments, enabling reproducible training runs, artifact versioning, and rollback capability that cut experiment turnaround from 3 days to 6 hours.
- **Engineered asynchronous ML inference APIs** using FastAPI, Celery, and Redis task queues, increasing throughput by 3x versus synchronous patterns and enabling non-blocking batch prediction for high-volume workflows.
- **Built CI/CD pipelines with GitHub Actions** for automated model validation, Docker image builds, and Kubernetes deployment, reducing manual deployment overhead by 80% and enabling zero-downtime rolling updates.
- **Developed React/Next.js dashboards** and Streamlit monitoring tools integrated with Prometheus and Grafana, giving stakeholders real-time visibility into model performance, feature drift, and A/B test results.
- **Architected a multi-database storage strategy** across PostgreSQL, MongoDB, and BigQuery, optimizing query plans for ML feature pipelines and reducing data retrieval latency by 40% for time-sensitive production models.

PROJECTS

Agentic AI Workflow for Regulatory Document Compliance

- **Architected a multi-agent compliance pipeline** using LangGraph on GCP Vertex AI, automating OCR-based extraction (Google Cloud Vision) and LLM-powered rule validation of FDA/USDA regulatory labels—reducing manual compliance review effort by an estimated 70% across a 10,000+ document corpus.
- **Engineered token-level prompt scaffolding and structured output tracing** to produce fully auditable justification chains for each compliance decision, enabling regulatory-grade explainability and cutting QA review cycles by 50%.

Clinical Document Intelligence Platform

- **Architected a healthcare-focused RAG platform** using GPT-4, LangChain, LlamaIndex, and FAISS to query clinical guidelines, patient care documents, and healthcare policy records—reducing manual document search time by an estimated 50%.
- **Engineered HIPAA-aware prompt handling**, citation-based responses, and structured output validation to improve retrieval accuracy by 35% and support auditable clinical decision-making.

AI Risk & Fraud Detection Assistant

- **Architected an LLM-powered fraud investigation assistant** using GPT-4, LangChain, FAISS, and anomaly detection models to automate transaction review, risk triage, and case summarization—reducing manual investigation time by an estimated 40%.
- **Engineered explainable risk scoring, human-in-the-loop review workflows**, and structured audit trails to improve fraud pattern detection, analyst decision support, and compliance-grade reporting.

Retail Inventory Intelligence — Kent Natural Food Store

- **Reduced inventory carrying costs by 25%** by building an end-to-end data analysis and visualization pipeline in Python and Tableau, identifying underperforming SKUs and modeling special-order strategies that optimized stock levels across 500+ product lines.

EDUCATION

Master of Science, Data Science — GPA: 3.881 /4.00 *December 2025*

Kent State University, Kent, OH